Is Non-IID an Artifact of Training from Scratch? A Critical Review of Data Heterogeneity in Federated Learning in the Pretraining Era

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***Abstract*—For nearly a decade, the field has treated non-identically-distributed (non-IID) client data as the defining obstacle of federated learning (FL). This framing motivated a large family of heterogeneity-specific algorithms. Researchers built that consensus almost entirely on models trained *from scratch*. A more recent body of evidence shows that when federated training instead begins from a *pretrained* (foundation-model) initialization, the gap between IID and non-IID training shrinks sharply. The gap narrows so far that heterogeneity-specific optimizers lose much of their advantage over a well-tuned FEDAVG/FEDADAM baseline. Yet the survey literature on non-IID FL, and the emerging literature on federated foundation models, continue to treat heterogeneity as a first-order *unsolved* problem. For parameter-efficient (LoRA) fine-tuning, feature shift, worst-client fairness, and out-of-distribution clients, heterogeneity in fact persists or is amplified. This review reconciles that contradiction. We make three contributions. First, we re-read five years of heterogeneity-specific methods as a single body of work. A from-scratch evaluation protocol confounds their reported gains. We then audit which gains survive a pretrained baseline. Second, we map the precise boundary between regimes where initialization *dissolves* the non-IID problem and regimes where it merely *relocates* it. Third, we propose an evaluation protocol under which the claim "pretraining solves heterogeneity" can be falsified. We argue the field should replace "design a better heterogeneity algorithm" with "report a pretrained-initialization baseline, quantify backbone-to-task alignment, and isolate the residual heterogeneity that pretraining cannot absorb."**

***Index Terms*—Federated learning, data heterogeneity, non-IID data, pre-training, foundation models, transfer learning, parameter-efficient fine-tuning, LoRA.**

I. INTRODUCTION

Federated learning [1] lets many clients train a shared model without centralizing their data. From its inception, the field identified *statistical heterogeneity* as the property that most sharply distinguishes FL from classical distributed optimization [2, 3]. Statistical heterogeneity means that clients hold local data that are not independent and identically distributed. Early work showed that non-IID data slows and destabilizes FEDAVG and can cost substantial accuracy [4–7]. A decade of algorithms followed to combat it. These include proximal and variance-reduced local updates [8–11], corrections for objective inconsistency [12], contrastive and dynamic regularizers [13, 14], adaptive server optimization [15], and personalization [16]. A standardized evaluation culture grew up alongside these methods. It includes Dirichlet label-skew partitions [17], naturally-partitioned benchmarks [18], and head-to-head studies [19, 20].

This entire edifice shares one rarely-stated assumption. **Training starts from random initialization.** Recent work has quietly undermined that assumption. Nguyen et al. [21] show that starting FL from a pretrained model makes the IID-vs-non-IID accuracy gap "significantly smaller." They also show that after pretraining, the choice of a heterogeneity-specific optimizer matters far less than the choice of server optimizer. Chen et al. [22] independently show pretraining "largely closes" the FedAvg-vs-centralized gap precisely in the hardest non-IID settings. Jhunjhunwala et al. [23] give a theoretical account. Data heterogeneity harms learning mainly through "misaligned filters," and pretraining leaves far fewer of them. Taken at face value, these results suggest that a large fraction of the non-IID literature was solving a problem that a good starting point substantially dissolves.

a) The contradiction this review addresses.: The field has not reconciled this. Every dedicated non-IID survey [24, 25] presupposes from-scratch training and catalogs heterogeneity-specific fixes. These surveys include the most recent and comprehensive one [26], which does not mention pretraining at all. At the same time, the surveys of federated foundation models [27–29] treat heterogeneity as a standing open challenge. Some report it is *worse* when fine-tuning foundation models. So two literatures coexist. One says "pretraining attenuates heterogeneity," and it comes from optimization papers, mostly in vision. The other says "heterogeneity is unsolved, perhaps worse," and it comes from surveys and federated-LLM work. No one has put them in the same frame.

b) Thesis.: Pretrained initialization substantially shrinks the label-skew IID gap for *full-model* federated fine-tuning on *in-distribution* metrics. In that regime it makes heterogeneity-specific optimizers largely unnecessary. But it does not eliminate heterogeneity. For parameter-efficient fine-tuning, feature/covariate shift, worst-client metrics, and out-of-distribution clients, heterogeneity persists or intensifies. Pretraining often *relocates* it into the aggregation operator or the classifier head. The non-IID problem is therefore neither "solved by pretraining" nor "unchanged by it." Its *locus and severity are a function of the starting point*. The field has not designed its benchmarks to see this.

c) Contributions.:

1) **A confound audit.** We treat "evaluated from scratch" as a threat to the validity of heterogeneity-specific FL methods. We then consolidate from published numbers how much

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of their reported advantage survives a pretrained baseline (Section IV).

2) **A boundary map.** We organize the evidence into the axes along which initialization does and does not dissolve heterogeneity. These axes are full-model vs. PEFT, shift type, metric, mechanism, and locus (Section V).
3) **A falsification protocol and agenda.** We specify the evaluation standard that could refute the thesis, and we reframe the research program around it (Section VII).

d) Scope.: The strongest evidence is vision-centric and label-skew. We foreground that. We treat multilingual/NLP results as corroboration. We treat federated LLM/LoRA fine-tuning as the principal *contested* regime rather than as settled either way. We conduct no new experiments. We draw all quantitative claims from the primary sources, and readers should check them against those sources. *(zero-experiment critical review)*

II. Background and the From-Scratch Consensus

The federated-learning literature has most consistently identified statistical heterogeneity as its central difficulty. The evidence that heterogeneity degrades learning appeared early. Researchers have rarely contested it. McMahan et al. [1] introduced FedAvg. They paired it with the pathological, sharply label-skewed client partitions that researchers still use to stress it. They already documented slower and less stable convergence as skew increases. Zhao et al. [4] supplied a mechanism. They traced the loss to the *weight divergence* that accumulates as locally trained models drift apart between synchronization rounds. They reported accuracy reductions of up to $\sim$55% in the extreme case where each client holds examples of only a single class. Li et al. [30] formalized the same phenomenon. They showed that heterogeneity enters directly into the rate at which FedAvg approaches a stationary point. Hsieh et al. [5] demonstrated that even ostensibly innocuous components become liabilities once clients hold non-identically distributed data. Batch-normalization layers are a particular example. By the time researchers wrote the field's consensus surveys, this picture had hardened into received wisdom. These surveys treated heterogeneity as a first-order, intrinsic obstacle of the federated setting [2, 3].

A. The algorithmic response to heterogeneity

The field framed the obstacle as intrinsic. So the response was overwhelmingly *algorithmic*. Researchers took the data as given and the architecture as fixed. That left two degrees of freedom for redesign. The first is how clients compute local updates. The second is how the server combines them. The resulting methods fall into a few recurring families. Table I maps them. The most direct, *data-level* family attacks the distributional gap itself. Zhao et al. [4] had diagnosed weight divergence. They also proposed distributing a small globally shared dataset. Each client then computes its local objective over a mixture closer to the global distribution. This trades a measure of privacy and communication for reduced skew. On CIFAR-10 they report recovering roughly 30% accuracy by globally sharing as little as 5% of the data. A second, *local-objective* family leaves the data untouched. It instead constrains the local update so that clients drift less far from the global model. FedProx adds a proximal penalty on deviation from the current global weights [8]. SCAFFOLD instead maintains control variates that estimate and subtract the client-drift direction. It corrects the update rather than merely penalizing it [9].

A third, *aggregation-level* family accepts heterogeneous local trajectories and repairs the combination step at the server. FedNova normalizes client contributions. This removes the objective inconsistency that arises when clients take unequal numbers of local steps [12]. Reddi et al. [15] recast server aggregation as an optimization problem in its own right. They applied adaptive optimizers such as FedAdam to the averaged update. FedDyn occupies an intermediate position. It adds a *dynamic* regularizer whose fixed point aligns each client's local stationary point with the global one. Exact local minimization then no longer pulls the global model off its optimum [14]. A final, *representation and personalization* family reasons about *what* a model learns rather than how. MOON aligns client and server representations through a model-contrastive penalty [13]. FedRep abandons the premise of a single shared model altogether. It learns a common feature extractor but equips each client with its own classifier head [16]. This was an early concession that accommodating some heterogeneity may work better than erasing it.

B. Evaluation culture and a hidden assumption

a) Synthesizing heterogeneity.: Researchers adjudicated these methods within a shared experimental culture. Naturally federated datasets are scarce. So they most often *synthesize* heterogeneity by partitioning a centralized benchmark. The Dirichlet label-partitioning scheme of Hsu et al. [17] became the de facto standard. Its concentration parameter α tunes skew continuously from mild to pathological. The naturally partitioned tasks of the LEAF suite complement it [18]. The most systematic head-to-head comparison is the NIID-Bench study of Li et al. [19]. It reached a deliberately sobering conclusion. Across partitioning schemes and datasets, *no single heterogeneity algorithm dominates*. The ranking of methods shifts from one skew regime to the next.

b) The hidden variable.: The field usually reads that non-result as a measure of how hard and how many-faceted heterogeneity is. That non-result admits a second reading. Every method and every benchmark above shares one modeling choice so ubiquitous that almost no one states it. In every case, the global model starts *from scratch*, from random initialization. Researchers vary and ablate partitioning schemes, local optimizers, aggregation rules, and architectures. They leave the starting point fixed. Initialization is thus a free variable that the from-scratch consensus never varied. The confound audit of Section IV and the evidence leading up to it show that varying it moves the conclusions. These results may describe heterogeneity *as such*. They may instead describe heterogeneity *as it appears from one particular and increasingly unrepresentative starting point*. The remainder of this review takes up that question.

TABLE I
A MAP OF THE FROM-SCRATCH ALGORITHMIC RESPONSE TO STATISTICAL HETEROGENEITY. EACH FAMILY INTERVENES AT A DIFFERENT POINT IN THE FEDERATED PIPELINE, BUT ALL TAKE THE GLOBAL MODEL TO BE *randomly initialized*. CITATIONS APPEAR IN THE TEXT.

Family	**Representative methods**	**Where it intervenes**
Data-level / shared data	globally shared or synthetic data	the client data distribution itself
Local-objective correction	FEDPROX, SCAFFOLD	the client's local optimization step
Aggregation-level	FEDNOVA, adaptive FEDOPT (FEDADAM)	the server's rule for combining updates
Dynamic regularization	FEDDYN	the local objective, varied across rounds
Representation & personalization	MOON, FEDREP	the learned representation / per-client parameters

III. THE ATTENUATION EVIDENCE

The from-scratch canon treats the IID-vs-non-IID gap as an intrinsic property of the federated learning problem. A second, more recent body of work complicates that reading. Once the shared model starts from a pretrained backbone rather than from random weights, the same gap attenuates sharply. Much of the specialized machinery built to close it then becomes difficult to distinguish from a carefully tuned baseline. We review this evidence along three strands. These strands are vision, language and multimodal transfer, and a nascent theory. We then surface a mechanism caveat that the evidence itself raises. That caveat motivates the boundary map of the next section. Throughout, we scope the claim deliberately. The attenuation we document here is strongest for *full-model* fine-tuning under *label*-distribution skew measured on *in-distribution* clients. We defer to the following section the regimes where it weakens, relocates, or reverses.

A. Vision

a) Optimizer choice washes out.: Nguyen et al. [21] fine-tune from pretrained initializations across CIFAR-10, FEMNIST, Stack Overflow, and Reddit. They report that the gap between IID and non-IID training shrinks substantially relative to training from scratch. Two secondary observations matter more for our purposes than the headline. First, the differences between federated optimizers contract. From a pretrained start, an adaptive *server* optimizer (FedAdam) contributes more than any heterogeneity-specific client correction. The choice of initialization also reorders the relative ranking of optimizers. Second, the study presents these conclusions through figures rather than through per-optimizer tables. We therefore read them as directional. We rely on the sign and ordering of the reported effects rather than on the single summary percentage quoted in the abstract, which the text does not operationally define. We also note that the study's optimizer suite does *not* include SCAFFOLD. That suite covers FedAvg, FedProx, FedAdam, FedNova, and MimeLite. So we can draw no SCAFFOLD-specific conclusion from it.

b) The gap shrinks most where it is largest.: Chen et al. [22] measure the gap between FedAvg and centralized training directly. They find that it collapses once the backbone starts from ImageNet pretraining, with the largest absolute improvement in the hardest settings (Table II). The partitions deserve precise statement, because secondary accounts frequently flatten them. Only CIFAR-10 uses a Dirichlet label split ($\alpha = 0.3$). Cityscapes uses a by-city partition and iNaturalist a geographic one. The latter two are naturalistic partitions rather than synthetic Dirichlet splits, and therefore plausibly combine label imbalance with feature shift. That the gap collapses even on these naturalistic splits is among the stronger pieces of evidence for the attenuation thesis.

c) Frozen features and closed-form heads.: A parallel line of work freezes the pretrained backbone and federates only a lightweight head. This work finds heterogeneity far easier to manage in feature space than in weight space. Legate et al. [31] show that a two-stage recipe improves over full fine-tuning by $+1.8$, $+10.4$, and $+6.1$ points on CIFAR-10, Flowers-102, and Stanford Cars, respectively (SqueezeNet backbone). That recipe is a federated nearest-class-mean head followed by fine-tuning (FedNCM+FT). The recipe cuts communication from roughly 120.25 GB to about 0.004 GB. They further observe that frozen-feature heads degrade more gracefully as label skew intensifies. Fanì et al. [32] push this to its limit with a closed-form ridge classifier on frozen features. A derivation shows that the estimator recovers the centralized solution and is invariant to client sampling. This derivation holds at full client participation, and the authors present it as such rather than as a formal theorem. Empirically, on iNaturalist it attains 45.1% accuracy *identically* across four different non-IID splits (standard deviation ± 0.0). As an initialization for subsequent full fine-tuning, it reaches 49.0% versus 39.5% without. It does so at up to roughly two orders of magnitude fewer communication rounds and correspondingly less compute.

B. Beyond Vision

The pattern extends beyond image classification. Weller et al. [33] report that pretraining narrows the non-IID text-classification gap to centralized performance from roughly 6.0 points when training from scratch to about 1.4 points when starting from a pretrained model. For machine translation, pretrained models *match or beat* centralized training. We phrase this cautiously, because the study includes no from-scratch machine-translation baseline against which to size the heterogeneity penalty. Tan et al. [34] extend the frozen-feature recipe across modalities by aggregating prototypes from mul-

TABLE II
THE FEDAVG-MINUS-CENTRALIZED PERFORMANCE GAP SHRINKS SHARPLY UNDER IMAGENET PRETRAINING [22]. EACH VALUE IS A GAP IN THAT BENCHMARK'S NATIVE METRIC. CIFAR-10 AND INATURALIST USE ACCURACY POINTS. CITYSCAPES USES MIOU POINTS. SMALLER IS BETTER. EACH PARTITION IS THE ONE THAT STUDY USES. THAT PARTITION IS *not* A DIRICHLET SPLIT FOR CITYSCAPES OR INATURALIST.

Benchmark	Partition	Metric	From scratch	Pretrained
CIFAR-10	Dirichlet ($\alpha = 0.3$)	Acc.	9.0	2.5
Cityscapes	by-city	mIoU	16.2	3.5
iNaturalist	geographic	Acc.	32.0	12.3

tiple frozen backbones. This suggests that the mechanism is not tied to a single architecture or input type.

C. A Supporting Theory

A theoretical account is beginning to explain *why* initialization should matter. Jhunjhunwala et al. [23] analyze two-layer convolutional networks. They argue that data heterogeneity does its damage primarily through *misaligned filters*. These are local solutions whose learned filters point in inconsistent directions across clients, so that averaging them is destructive. Pretraining, on this account, supplies a shared, well-aligned filter basis. It thereby leaves fewer such misaligned directions for heterogeneity to exploit. The mechanism is consistent with the empirical finding that the benefit is largest precisely where from-scratch training would otherwise diverge most.

D. A Mechanism Caveat

The same body of work that documents the attenuation also circumscribes it. Chen et al. [22] are explicit that pretraining stabilizes *global aggregation*. It makes the averaged model well behaved. But it does *not* reduce *local client drift*. Clients still move apart during local training. On this reading the gap narrows for a specific reason. It narrows not because pretraining removes heterogeneity from the optimization, but because a shared initialization makes the divergent local updates far more compatible under averaging. The distinction is easy to lose when the only reported metric is in-distribution global accuracy. This distinction is the hinge on which the remainder of this review turns. Suppose attenuation is a property of the *aggregation operator* meeting a good initialization, rather than of the heterogeneity itself. Then changing what one aggregates may *relocate* the penalty rather than eliminate it. Examples of such a change include a low-rank adapter instead of a full model, a classifier head in isolation, or a client with genuinely shifted features. The next section maps where that relocation occurs.

IV. A CONFOUND AUDIT OF WHETHER THE REPORTED GAINS SURVIVE A PRETRAINED BASELINE

Researchers designed, tuned, and reported nearly every heterogeneity-specific method surveyed in Section II against a *from-scratch* FEDAVG baseline. This makes initialization a silent confound. A method's headline advantage is the distance between two randomly-initialized training runs. One run uses the proposed correction. The other does not. That distance need not survive when the shared starting point moves. Suppose a pretrained initialization alone closes most of the IID-vs-non-IID gap that a method set out to close. Then the method's *marginal* contribution over a well-tuned baseline may be small or absent. On some metrics it may even be negative. This audit is the first of our three contributions. It asks from published numbers alone how much of the reported advantage is a property of the algorithm and how much is an artifact of the from-scratch protocol. Table III consolidates the cases in which researchers measured the same or a closely comparable setting from both a from-scratch and a pretrained start. We read it here. We stay deliberately explicit about what it cannot show.

A. Two patterns in the cross-initialization evidence

a) Attenuation and compression under global, in-distribution label skew.: The upper rows of Table III are mutually consistent. They point in one direction. Chen et al. [22] report that the FEDAVG-minus-centralized gap under non-IID data shrinks from 9.0 to 2.5 points on CIFAR-10 (Dirichlet α=0.3). It shrinks from 16.2 to 3.5 mIoU on a by-city split of Cityscapes. It shrinks from 32.0 to 12.3 on a geographically partitioned iNaturalist. The benefit is *largest in the hardest settings*. Nguyen et al. [21] observe two things. First, the IID-vs-non-IID gap becomes much smaller from a pretrained start. Second, the *differences among federated optimizers compress*. An adaptive *server* optimizer (FEDADAM) matters more than any heterogeneity-specific local correction. Initialization also reorders the ranking of optimizers. These comparisons appear graphically rather than as a per-optimizer table. This compression is the audit's central object. It is precisely the inter-method gap that the from-scratch literature reported as each method's contribution. Two further results push the attenuated gap toward zero. On frozen features, Fanì et al. [32] obtain 45.1% on iNaturalist. This value is *identical* (standard deviation ± 0.0) across four different non-IID partitions. Their method is a closed-form ridge classifier. A derivation at full client participation, not a formal theorem, shows it to equal the centralized solution and to stay invariant to client sampling. The heterogeneity penalty is therefore zero by construction rather than by tuning. A two-stage frozen-feature-then-fine-tune recipe likewise improves over full fine-tuning (by $+1.8$, $+10.4$, and $+6.1$ points on CIFAR-10, Flowers-102, and Stanford Cars) while degrading more gracefully under skew [31]. The pattern is not confined to vision. In multilingual text classification the non-IID gap to centralized falls from roughly 6.0 to 1.4 points once a pretrained encoder is in use [33]. Across these rows, a shared and strong starting point does not overturn the from-scratch head-to-head finding that *no single heterogeneity algorithm dominates* [19]. It renders

TABLE III
INITIALIZATION CHANGES THE NON-IID PICTURE. WE VERIFIED ALL VALUES AGAINST THE PRIMARY SOURCES. THE LAST COLUMN GIVES THE LOCATION OF EACH VALUE. THE FINAL ROW COMES FROM A NON-PEER-REVIEWED PREPRINT. "GAP" MEANS REPORTED DEGRADATION ATTRIBUTABLE TO NON-IID OR TO FEDERATION UNDER NON-IID (PERCENTAGE POINTS UNLESS NOTED). THE AUDIT IS NECESSARILY PARTIAL. FEW STUDIES REPORT THE *same* HETEROGENEITY METHOD UNDER BOTH A FROM-SCRATCH AND A PRETRAINED BASELINE. THAT ABSENCE IS ITSELF THE EVIDENCE GAP MOTIVATING SECTION VII.

Source / setting	Metric	From scratch	Pretrained	Location
Chen et al. [22], CIFAR-10 (α=0.3)	FedAvg−centralized gap	9.0	2.5	Table 2
Chen et al. [22], Cityscapes (by-city)	gap (mIoU)	16.2	3.5	Table 5
Chen et al. [22], iNaturalist (geo)	gap	32.0	12.3	Table 4
Weller et al. [33], text cls. (per-lang.)	gap to centralized	$\sim$6.0	$\sim$1.4	Table 3
Fanì et al. [32], iNat (4 non-IID splits)	acc. (split variance)	*(n/a)*	45.1 ($\pm$0.0)	Table 1
Nguyen et al. [21], multi-optimizer	het.-opt. edge over FedAvg	positive	$\approx$0	Figs. 1,4
Nguyen et al. [21], CIFAR-10 (α=0.1), FedAdam	acc. (random vs. pretrained init)	50.6	75.9	Table 9 (app.)
Naseer and Shoaib [35], α=0.1 *(preprint)*	**worst-client acc. gap**	32.2%	50.1%	Table 3

that finding moot. When the starting point is shared and strong, far less inter-method gap remains for any method to dominate.

b) A worst-client sign flip.: The final row of Table III runs the other way. In a non-peer-reviewed preprint, Naseer and Shoaib [35] report a *worst-client accuracy gap* at extreme skew (α=0.1). This gap measures the shortfall of the least-served client, not an accuracy. It is 32.2 points for a from-scratch TextCNN. It rises to 50.1 points for a pretrained DistilBERT fine-tuned with LoRA. Here the pretrained foundation model *amplifies* disparity rather than attenuating it. The same class of initialization that compresses the global-accuracy gap widens the tail. We flag this result as a preprint and as a comparison of *gaps* rather than accuracies. It also couples a foundation model to parameter-efficient fine-tuning. That is the regime whose aggregation pathology we dissect in Section V. This result reads best as a boundary case rather than as a clean like-for-like inversion of the upper rows.

B. What the audit can and cannot establish

This audit has an honest limitation. *Very few papers report both initializations for the same method*. Published numbers therefore yield only a partial method-by-method ledger. The strongest cross-initialization evidence concerns the *envelope* rather than any single named algorithm. That envelope is the IID-vs-non-IID gap and the spread among optimizers. Where the evidence covers named methods at all, that coverage is thin and graphical. Nguyen et al. [21] vary initialization across FEDAVG, FEDPROX [8], FEDNOVA, MimeLite, and FEDADAM, and present the result only in figures. Canonical corrections such as SCAFFOLD [9], MOON [13], and FEDDYN [14] do not appear under a matched pretrained baseline at all. The published record does not permit a fully method-by-method audit. Such an audit would re-evaluate Li et al. [8], Karimireddy et al. [9], Li et al. [13], and Acar et al. [14] each from a *matched* pretrained start. Their authors and successors almost never reran these methods from a pretrained initialization. That absence is not a gap in this review but a finding about the literature. The single most decisive experiment for the heterogeneity-algorithm program is a matched from-scratch-versus-pretrained comparison of each method. Researchers have largely not run it. The clearest available instance is the per-optimizer accuracy of Nguyen et al. [21] (Table III). There FEDADAM on CIFAR-10 at α=0.1 rises from 50.6 to 75.9 with a pretrained start. Even there, Nguyen et al. [21] report the *relative* ordering of optimizers only graphically. That ordering is the quantity that would reveal whether a method's advantage survives.

On a conservative reading, the audit neither supports the claim that pretraining *solves* non-IID nor amounts to a per-method refutation of the heterogeneity literature. What it does support is narrower and sharper. Consider global, in-distribution accuracy under label skew with full-model fine-tuning. There a shared pretrained start shrinks the non-IID penalty. It also compresses the inter-method differences that motivated much of that literature. Heterogeneity-specific optimizers are therefore largely redundant in that regime. On a worst-client metric under extreme skew, by contrast, the same start can instead widen the gap. *Which* heterogeneity problem one observes is therefore a function of initialization, metric, and regime. The boundary map of Section V makes that dependence systematic.

V. THE BOUNDARY MAP OF WHERE HETEROGENEITY PERSISTS OR INVERTS

Researchers have by now established the following for *full-model* fine-tuning from a foundation-model initialization under *label*-distribution skew. The IID-vs-non-IID accuracy gap shrinks substantially on *in-distribution* metrics. Heterogeneity-specific optimizers also lose much of their advantage over

a well-tuned server optimizer in that regime [21, 22, 33]. One finding is emblematic. The CIFAR-10 FedAvg-minus-centralized gap falls from 9.0 to 2.5 points under Dirichlet α=0.3 with ImageNet pretraining [22]. Our confound audit (C1) finds that such gains largely survive a strong pretrained baseline.

This dissolution holds only within a particular operating regime. That regime combines full-model adaptation, label-distribution skew, average in-distribution accuracy, short local training in which aggregation dominates, and a backbone whose pretraining domain matches the task. Each condition is a degree of freedom that standard federated benchmarks silently fix. These benchmarks fix Dirichlet label partitions [17], averaged accuracy, and modest local computation over from-scratch or ImageNet-pretrained models [1, 19]. In this section we vary each degree of freedom in turn and ask where the gap returns. The recurring answer is that heterogeneity seldom disappears. Instead it *relocates*, typically into the aggregation operator or the classifier head. It also *rescopes* from the average client to the worst client and from in-distribution to unseen clients. We organize these findings into five axes, summarized in Table IV.

A. Five axes along which initialization stops helping

a) Adaptation method, full fine-tuning versus parameter-efficient adaptation.: The dissolution result applies specifically to updating all weights. Parameter-efficient adaptation is the default for large pretrained models. When adaptation is parameter-efficient, heterogeneity re-enters through the *structure* of aggregation rather than the data. In LoRA, each client learns a low-rank update $\Delta W = BA$. FedAvg averages A and B separately. The server then realizes $\mathrm{mean}(B)\,\mathrm{mean}(A)$, whereas the aggregate of the local products is $\mathrm{mean}(BA)$. Sun et al. [36] show, for the two-client equal-weight case, that these are unequal. They also show that multi-step local updates and differential-privacy noise compound the discrepancy. Notably they attribute the instability to heterogeneity and local updating, *not* to pretraining. The pretrained start does not rescue LoRA aggregation. Freezing A and training only B restores linearity ($\mathrm{mean}(B)\,A = \mathrm{mean}(BA)$ when A is shared) and halves per-round communication at fixed rank. Exact or corrected aggregation schemes target the same non-linearity directly [37, 38]. The problem sharpens when clients adopt *different* ranks. Aggregating heterogeneous-rank updates is ill-posed and biases the global model toward larger-rank clients [39, 40]. Reports show that federated LoRA continues to struggle under the FedAvg paradigm even for recent methods (a preprint, 41). The non-IID difficulty here is not a property of the client data distributions *per se*. It is a property of an averaging operator that is no longer linear in the quantity of interest.

b) Type of distribution shift.: Essentially all of the dissolution evidence concerns label-distribution skew. Other shift types define distinct axes. On these axes, no evidence shows that initialization helps. Under feature/covariate shift, Li et al. [42] show that reconciling divergent input distributions requires client-local batch-normalization statistics. This need is orthogonal to the backbone initialization, because it concerns per-client normalization rather than weight alignment. Under test-time or unseen-client shift, Yang et al. [43] find that a single global adapter is insufficient even atop a foundation model. Robust performance requires *both* a global and a client-local adapter. These results indicate that the gap-shrinkage for label skew does not transfer automatically to feature or concept shift. No controlled study isolates shift types while holding a *matched* pretrained backbone fixed. Such a study would separate label, feature/covariate, concept, and quantity skew. The claim that initialization helps less for non-label shifts therefore rests on inference across heterogeneous studies. No single setup measures it. We convert this evidence gap into a reporting requirement in Section VII.

c) Choice of evaluation metric.: Average accuracy can improve while the client-level tail deteriorates. In a preprint, Naseer and Shoaib [35] report that at extreme skew (α=0.1) the *worst-client accuracy gap* is 32.2 points for a from-scratch TextCNN but 50.1 points for a pretrained DistilBERT adapted with LoRA. In other words, the foundation model *amplifies* worst-client disparity. (These are gaps, not accuracies, from a non-peer-reviewed source. We treat the direction, not the exact magnitude, as indicative.) The effect compounds with the first axis, because the pretrained condition also uses LoRA and thus inherits the same aggregation non-linearity. The broader point is simple. A benchmark reporting a single averaged number cannot see a fairness regression. A method that tightens the mean can loosen the tail.

d) Mechanism, global aggregation versus local drift.: Where in the training dynamics does initialization act? Chen et al. [22] are explicit on this point. Pretraining stabilizes *global aggregation*, so the averaged model is better behaved. It does *not* reduce *local* client drift during on-device training. The theory of Jhunjhunwala et al. [23] is consistent. For two-layer CNNs, heterogeneity harms chiefly through *misaligned filters*, and a pretrained start simply begins with fewer of them. Both accounts locate the benefit at initialization and in the aggregation step, not in the local trajectory. This yields a falsifiable prediction. Local computation can grow through many local epochs, long round budgets, or task distributions far from the pretraining corpus. As it grows, accumulated local drift should overwhelm the initial alignment. The IID-vs-non-IID gap should then re-emerge even from a strong start. No study sweeps the local-epoch count or total round budget against a matched pretrained initialization. Such a sweep would test whether the dissolution of the gap is transient and re-emerges as local drift accumulates, rather than permanent. This prediction follows directly from the aggregation-versus-drift mechanism of Chen et al. [22] and remains untested. We therefore elevate it to a falsification criterion in Section VII.

e) Locus within the network.: Even when the backbone is strong or frozen, residual heterogeneity concentrates in the classifier head. Li et al. [44] show that under label skew the head remains bias-prone atop a strong, frozen backbone. The feature extractor may be well-aligned while the linear decision layer still encodes each client's local label marginal. This localizes the problem. A derivation at full client participation shows that a closed-form head on frozen features recovers the

centralized solution and stays invariant to client sampling [32]. That is precisely what one expects if the head is where label skew lives. A second locus effect concerns *which* pretraining. Yan et al. [45] find that under severe heterogeneity, in-domain self-supervised pretraining outperforms generic ImageNet pretraining. The benefit tracks *domain alignment* between the pretraining corpus and the federated task, not the mere fact of pretraining. Initialization therefore helps to the extent that it supplies features already aligned to the clients' data. Where it does not, heterogeneity persists in the head and in the feature mismatch.

B. Synthesis of relocation and rescoping, not removal

Read together, the five axes show that pretraining does not so much solve data heterogeneity as change where it lives and how we measure it. Along the adaptation axis it moves into the aggregation operator. Along the locus axis it moves into the classifier head. Along the mechanism axis it shifts to the local trajectory rather than disappearing. Along the metric axis it migrates from the mean to the tail. Along the shift-type axis it simply stays intact for non-label shifts. The common thread is that the dissolution that the literature reports is real but regime-bound. It holds at the intersection of full fine-tuning, label skew, average accuracy, short local training, and in-domain data. That intersection is almost exactly the operating point that standard benchmarks instantiate [1, 17, 19]. The locus and severity of non-IID thus depend on the starting point. Each axis is a separate failure surface. So reporting a single averaged in-distribution number from a pretrained full-fine-tuning run systematically hides four of the five. Table IV therefore also specifies what a benchmark would have to vary in order to *see* residual heterogeneity. That is the subject of the falsification protocol that follows.

VI. Reconciliation

Taken together, the two lines of evidence reviewed above read as a contradiction. The first reports that a pretrained initialization shrinks the IID-versus-non-IID gap to near-triviality and makes heterogeneity-specific optimizers largely redundant. The second reports that heterogeneity remains unsolved under federated foundation-model fine-tuning, and in places is amplified. We argue that these lines do not disagree about any fact. They disagree about what each study held fixed while measuring something else. Each result is faithful to its own *evaluation regime*. The regimes differ along four axes that benchmarks inherited from the from-scratch era leave implicit. The first axis is the fine-tuning surface (full-model versus parameter-efficient). The second is the kind of distributional shift (label skew versus feature/covariate shift versus out-of-distribution clients). The third is the reported metric (global in-distribution accuracy versus worst-client outcomes). The fourth is the optimization horizon (few versus many local steps). Once these axes are named, the apparent contradiction resolves into a partition of the regime space rather than a dispute over results.

a) The attenuation results occupy one corner of that space.: The optimization line measures global, in-distribution accuracy under label-distribution skew with *full-model* fine-tuning. In that corner the gap genuinely attenuates. Chen et al. [22] report the FedAvg-to-centralized accuracy gap falling from 9.0 to 2.5 points on CIFAR-10 at $\alpha{=}0.3$, with the largest absolute reductions on the hardest natural splits. Nguyen et al. [21] study vision and language benchmarks. They find that the IID/non-IID gap becomes much smaller from a pretrained start, that the spread between federated optimizers collapses, and that an adaptive *server* optimizer matters more for the remaining gap than any heterogeneity-specific client correction. Jhunjhunwala et al. [23] supply a mechanism. They argue for two-layer CNNs that heterogeneity harms mainly through misaligned filters and that a pretrained start leaves fewer of them. The same corner holds in text classification, where Weller et al. [33] report the non-IID gap to centralized shrinking from roughly 6.0 to 1.4 points. Crucially, attenuation is partial even here. The hardest split of Chen et al. [22] retains a 12.3-point gap after pretraining (down from 32.0 on a geographic partition). Those authors note that pretraining stabilizes *global aggregation* without reducing *local client drift*. Pretraining has not removed the heterogeneity. The global in-distribution metric has stopped registering it.

b) The persistence results occupy the rest.: Move one axis and the attenuation claim lapses. Change the fine-tuning surface to a parameter-efficient one and the difficulty reappears inside the aggregation operator. Averaging the low-rank LoRA factors separately is biased because $\mathrm{mean}(B)\,\mathrm{mean}(A) \neq \mathrm{mean}(BA)$ (shown for the two-client equal-weight case) [36]. Exact and corrected aggregation schemes confirm this defect by removing it [37, 38]. Heterogeneous client ranks compound it by biasing the result toward larger ranks [39, 40]. The authors attribute the instability to heterogeneity, multi-step local updates, and differential-privacy noise, *not* to the pretrained start [36]. Federated LoRA continues to struggle with heterogeneity under the FedAvg paradigm more broadly [41]. Change the shift from label to feature/covariate and each client must keep local normalization statistics regardless of initialization [42]. Change the metric from the global mean to the worst client and the comparison can invert. A preprint (not peer-reviewed) reports that at $\alpha{=}0.1$ the worst-client accuracy *gap* widens from 32.2 points for a from-scratch TextCNN to 50.1 points for a pretrained DistilBERT with LoRA [35]. Change the clients from in-distribution to out-of-distribution or test-time and a client needs both a global and a local adapter even atop a foundation model [43]. Even under label skew, the classifier head remains bias-prone atop a strong or frozen backbone [44]. The federated-foundation-model surveys that do inspect this regime concur that heterogeneity is as hard or harder under FM fine-tuning [28, 29].

c) The contradiction is an artifact of under-specified regimes, and the surveys show why it has persisted.: No fact is in tension. What is in tension is a pair of unstated quantifiers. The attenuation claim is implicitly *full-model, label-skew, global-accuracy, short-horizon*. The persistence claim is *everything else*. That this has gone unreconciled is structural rather

TABLE IV
THE BOUNDARY MAP. PRETRAINING DISSOLVES THE IID-VS-NON-IID GAP ONLY WITHIN THE REGIME IN THE SECOND COLUMN. VARYING ANY AXIS RELOCATES OR AMPLIFIES HETEROGENEITY (THIRD COLUMN). CITATIONS GIVE REPRESENTATIVE EVIDENCE. FULL SUPPORT APPEARS IN THE TEXT.

Axis	Gap dissolves when…	Heterogeneity persists / inverts when…	Evidence
Adaptation	full-model fine-tuning	PEFT/LoRA, with non-linear aggregation and heterogeneous ranks	[36, 39]
Shift type	label-distribution skew	feature/covariate or unseen-client shift	[42, 43]
Metric	global / average accuracy	worst-client (fairness) tail	[35]
Mechanism	global aggregation dominates	local drift dominates (long local training, OOD tasks)	[22, 23]
Locus	backbone features aligned to task	classifier head, or mismatch between pretraining and task domains	[44, 45]

than empirical. The surveys that frame the gap as intrinsic fall into two groups. Some predate the pretraining results [3, 24]. Others are current yet decline to treat initialization as a variable at all. Initialization is absent from the taxonomy, body, and future-directions of the most thorough recent non-IID survey [26]. It is absent too from earlier algorithmic treatments that enumerate only from-scratch fixes [25]. The one survey that explicitly names the small-from-scratch versus pretrained-FM mismatch stops at its efficiency implications [46]. Holding none of the regime axes fixed, a survey cannot help but read the two lines as a disagreement. Making the axes explicit is precisely our boundary map (contribution C2). The map also tells the confound audit (C1) which baseline a reported gain must survive. It tells the falsification protocol (C3) where a positive result would be most surprising. One caveat the map must carry is that the width of the attenuation corner is not a constant of pretraining but of *alignment*. Under severe skew, in-domain self-supervised pretraining can beat generic ImageNet pretraining. So the benefit tracks domain match rather than pretraining as such [45].

A. The boundary map

We summarize the partition in Figure 1, the review's centerpiece and the realization of contribution C2. Each cell's region assignment is a conceptual synthesis of the evidence in Sections III–V. We read it from the cited studies' reported settings (dataset, partition, metric, and optimization horizon), not from a transcription of any single table.

VII. A FALSIFICATION PROTOCOL AND RESEARCH AGENDA

The confound audit (Section IV) and the boundary map (Section V) establish two things. A strong initialization does not make heterogeneity disappear so much as *relocate* it. The field's benchmarks were not built to see where. A thesis of that shape is useful only if it can be refuted. This section delivers the review's third contribution. We give a reporting and falsification protocol that makes the thesis testable. We also give a research agenda. The agenda redirects effort from inventing heterogeneity algorithms toward characterizing the heterogeneity that a good starting point leaves behind.

A. A Reporting and Falsification Protocol

Stated precisely, the thesis makes two predictions. Each one is refutable. The first prediction is *dissolution*. Consider full-model fine-tuning under label-distribution skew, evaluated on in-distribution metrics. There a well-tuned pretrained FEDAVG/FEDADAM baseline shrinks the IID-vs-non-IID gap enough that heterogeneity-specific optimizers are largely unnecessary. So a proposed method's advantage should *not* survive substitution of such a baseline. An adaptive *server* optimizer should absorb most of what remains [21]. The second prediction is *relocation*. It concerns settings outside that envelope. These include parameter-efficient fine-tuning, feature/covariate shift, worst-client metrics, and out-of-distribution clients. In these settings the gap persists or widens. It then reappears in the aggregation operator or the classifier head. Table V turns these predictions into five reporting requirements. Each one targets a specific way current practice can conflate initialization with algorithm, or let relocated heterogeneity go unmeasured.

The protocol is falsificationist in the strict sense. It names what would refute each half of the thesis. A single clean counterexample refutes the dissolution prediction. That counterexample is a label-skew, full-model, in-distribution setting with a well-tuned pretrained baseline in which a heterogeneity-specific optimizer still wins by a margin that survives requirement (i). The converse refutes the relocation prediction. One would have to show that a pretrained full-model baseline already closes the worst-client (ii) and out-of-distribution (iii) gaps, with no residual advantage for the client-local and aggregation-level corrections that presently close them [42–44]. Making both halves refutable is the point. It converts a narrative about pretraining into measurements any future benchmark can run.

B. A Research Agenda, from Inventing Algorithms to Characterizing Residual Heterogeneity

The from-scratch era answered heterogeneity by proposing optimizers, and the controlled verdict was that none dominates [19]. If a strong initialization dissolves the gap in the common regime and merely relocates it in the others, the productive question is no longer *which new algorithm* but *where, and how*

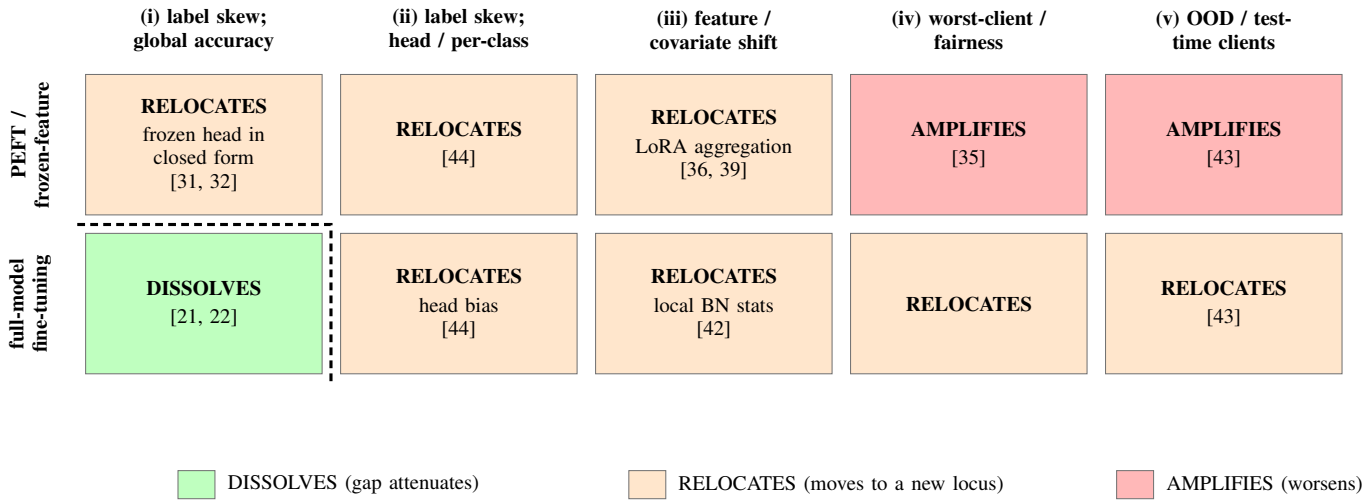


Fig. 1. **The boundary map, showing where a pretrained initialization dissolves, relocates, or amplifies the effect of non-IID data.** *DISSOLVES* (lower-left) covers full-model fine-tuning under label skew, read off global in-distribution accuracy. There the IID/non-IID gap attenuates [21, 22, 33] for the misaligned-filter reason given by Jhunjhunwala et al. [23]. *RELOCATES* (center band) is where the gap does not close but moves to a new locus. It moves into client-local normalization statistics under feature/covariate shift [42]. It moves into the classifier head under label skew [44], which the frozen-feature recipes turn to advantage by solving the head in closed form on fixed features [31, 32] and by aggregating frozen-backbone prototypes across modalities [34]. For parameter-efficient fine-tuning it moves into the aggregation operator, where separate averaging of low-rank factors is biased [36–38], heterogeneous ranks worsen the bias [39, 40], and the difficulty persists under FedAvg more broadly [41]. *AMPLIFIES* (upper-right) covers worst-client disparity at extreme skew [35] and out-of-distribution / test-time clients [43], matching federated-foundation-model surveys that report heterogeneity worsening under FM fine-tuning [28, 29]. The DISSOLVES/RELOCATES boundary traces the observation of Chen et al. [22] that a pretrained start stabilizes global aggregation without reducing local client drift. Left of it the stabilized global metric no longer registers the heterogeneity. Right of it the drift resurfaces in a different operator or metric. The horizontal extent of DISSOLVES is contingent on alignment between the pretraining corpus and client domains [45], not on pretraining alone.

TABLE V
A REPORTING AND FALSIFICATION PROTOCOL FOR STUDIES OF DATA HETEROGENEITY IN FEDERATED LEARNING. EACH REQUIREMENT TARGETS A SPECIFIC WAY CURRENT PRACTICE CAN CONFLATE INITIALIZATION WITH ALGORITHM, OR LEAVE RELOCATED HETEROGENEITY UNMEASURED.

	Report	**What it tests / guards against**
(i)	A pretrained-initialization baseline beside any from-scratch result, and whether a proposed method's advantage *survives* it	Confounding initialization with the heterogeneity algorithm. From scratch "no algorithm dominates" [19], yet from a pretrained start the optimizer gap compresses [21, 22]
(ii)	Per-client and worst-client accuracy curves, not only global accuracy	Global means that hide tail disparity under skew, or even invert it [35, 44]
(iii)	At least one feature/covariate-shift partition and one out-of-distribution / unseen-client partition, not only label skew	Label-skew-only benchmarks that cannot see where a frozen backbone fails [18, 42, 43]
(iv)	A full-model vs. parameter-efficient (e.g. LoRA) ablation whenever a foundation model is used	PEFT silently relocating heterogeneity into the aggregation operator [28, 36]
(v)	A measure of backbone–task alignment for the chosen initialization	Crediting "pretraining" with a benefit that in fact tracks distributional proximity [45]

much, heterogeneity remains once the starting point is fixed. We promote five open problems in place of a sixth algorithm.

a) Skew-type decomposition.: The pro-attenuation evidence is almost entirely Dirichlet label skew [17]. A decomposition that isolates label, feature/covariate, and quantity skew under a *matched* pretrained backbone would reveal which components initialization absorbs and which it does not. The boundary map already indicates that feature shift is not absorbed and still demands client-local statistics [42].

b) A drift-vs-aggregation scaling law.: Chen et al. [22] report that pretraining stabilizes *global aggregation* without reducing *local* client drift. This invites a scaling law for the residual gap as a joint function of local epochs, skew, and backbone–task alignment, $(E, \alpha, \text{align})$. Such a law would identify the regime in which drift, rather than aggregation noise, dominates. In that regime a heterogeneity-specific method is therefore still warranted.

c) Does exact aggregation remove the need for heterogeneity-specific optimization?: Federated LoRA is unstable under heterogeneity. The instability arises because the server averages the low-rank factors separately, and $\text{mean}(B)\,\text{mean}(A) \neq \text{mean}(BA)$. Multi-step local updates and differential-privacy noise compound it [36]. Exact and corrected aggregation now exists in closed form for the linear head [32] and is emerging for LoRA [37, 38]. One open question remains. Does exact aggregation do for parameter-

efficient fine-tuning what a pretrained baseline did for full fine-tuning, rendering heterogeneity-specific optimizers largely unnecessary? Or do heterogeneous client ranks reintroduce the problem? Aggregating unequal ranks is ill-defined and biased toward larger-rank clients [39, 40].

d) The worst-client sign-flip threshold.: A preprint reports that at $\alpha{=}0.1$ a pretrained model with LoRA exhibits a *larger* worst-client accuracy *gap* than a from-scratch baseline (50.1 vs. 32.2 points). That is, the foundation model *amplifies* worst-client disparity at extreme skew [35]. Read this as a single data point, entangled with PEFT. It motivates locating the skew threshold $\alpha^\star$ at which the sign of initialization's effect on worst-client disparity flips from helpful to harmful. The search would vary initialization and PEFT jointly. Fairness then becomes a design constraint rather than an afterthought.

e) An ex-ante *alignment predictor.:* Under severe heterogeneity, in-domain self-supervised pretraining outperforms generic pretraining, so the benefit tracks backbone–task alignment rather than pretraining per se [45]. The agenda's capstone is a predictor, computable *before* training, of when the IID-vs-non-IID gap returns. Such a predictor would tell a practitioner in advance whether a strong initialization will suffice or a heterogeneity-specific method remains necessary. A standardized, training-free measure of backbone–task alignment of the kind this predictor (and requirement (v)) presupposes does not yet exist in the federated literature. To our knowledge no standard, training-free backbone–task alignment metric is yet established in the federated-learning literature. One practical candidate is feature-space class separability on a small shared probe set. An example is a silhouette or Fisher-ratio score on frozen features. Such a score needs neither training nor raw-data exchange. It could serve as a cheap ex-ante predictor of whether the gap will return.

None of this asserts that heterogeneity is solved. It asserts that the *locus and severity* of non-IID are contingent on the starting point. The surveys that still omit initialization from their taxonomy of heterogeneity [26] must be rebuilt to measure it. So must the surveys that name the small-from-scratch/pretrained-foundation-model mismatch but stop at its efficiency implications [46].

VIII. Positioning Relative to Existing Surveys

No existing survey frames the non-IID gap as *initialization-dependent*. The literature that bears on heterogeneity has instead settled into two camps that do not speak to each other. One is a heterogeneity-survey tradition that predates or sets aside pretraining. It treats the IID-vs-non-IID gap as an intrinsic property of the client partition. The other is an emerging federated-foundation-model literature. It registers heterogeneity as a standing obstacle under foundation-model fine-tuning. It never measures that obstacle against the evidence that a pretrained start attenuates it. Neither camp puts the two bodies of evidence in a single frame. That reconciliation is the gap this review occupies. The review fills it through the confound audit of Section IV, the boundary map of Section V, and the falsification protocol of Section VII.

A. Surveys that treat the gap as intrinsic

a) The pre-pretraining consensus.: Zhu et al. [24] and Kairouz et al. [3] predate the attenuation results. They treat the heterogeneity gap as a given. They catalog its causes and the algorithmic remedies raised against it. Yet the initialization of the model never enters as a variable. This reflects their moment rather than an oversight. No one had yet run the experiments that would reframe the gap as regime-dependent. Still, the consensus they codified cannot, on its own terms, anticipate that a pretrained start would substantially shrink the gap for full-model fine-tuning under label skew.

b) Algorithmic-remedy taxonomies.: Ye et al. [25] organizes the field around from-scratch algorithmic interventions. It contains no category for the choice of initialization. A reader of that taxonomy would conclude that heterogeneity calls for *modifying the optimizer or the aggregation rule*. That is precisely the class of method whose necessity the attenuation evidence calls into question for the full-model, label-skew, in-distribution regime (Section III).

c) The most thorough recent synthesis.: The sharpest instance of the blind spot is Jimenez Gutierrez et al. [26], the most current and comprehensive non-IID survey available. Pretraining, foundation models, and initialization appear nowhere in its body, its taxonomy, or its future-directions. The terms surface only *inside the titles of two cited papers*, never as objects of discussion. The field's most thorough recent survey can treat heterogeneity at length without registering initialization as a factor. That is the clearest sign that the gap's init-dependence has not yet entered the field's self-description. It is the key positioning target for the present review. The measurement apparatus partly explains the blind spot. These surveys rest on a benchmark culture of Dirichlet label-skew partitions and controlled studies that found no single heterogeneity algorithm dominant [17, 19], none of which varies the starting point.

B. Surveys that treat heterogeneity as a persisting foundation-model challenge

On the other side of the divide, the surveys of federated foundation models do engage pretrained models. They treat them only as a setting in which heterogeneity survives. They never present the move to pretraining as something that can also reduce the severity of that heterogeneity. Ren et al. [27], Li et al. [28], and Fan et al. [29] all treat non-IID as a persisting, largely unsolved challenge. Li et al. [28] in fact reports that heterogeneity is *worse* when fine-tuning foundation models. Fan et al. [29] lists it among the open problems of the area. None of them sets this against the attenuation evidence in vision [22] and language [21, 33]. Heterogeneity shrinks in one set of experiments and grows in another. These surveys therefore never name that apparent contradiction, let alone resolve it. We argue (Section VI) that the contradiction is only apparent. Attenuation and amplification describe different regimes. Attenuation covers full-model fine-tuning under label skew on in-distribution metrics. Amplification covers parameter-efficient fine-tuning, feature shift, worst-client outcomes, and out-of-distribution clients.

Separating the two regimes is the work of the boundary map (Section V).

a) A survey that names the mismatch.: The closest prior work to our framing is Woisetschläger et al. [46]. It explicitly identifies the mismatch between the small, from-scratch models of the classical heterogeneity literature and the pretrained foundation models of current practice. It stops, however, at the efficiency and systems consequences of that mismatch. It notes that the setting has changed. But it never asks what the change does to the IID-vs-non-IID gap or to the heterogeneity-specific methods built to close it.

C. A coverage map

Table VI summarizes the divide along the four capabilities a reconciling survey would need. The first is coverage of the classical from-scratch non-IID literature. The second is coverage of the pretraining-attenuation effect. The third is an explicit reconciliation of the two. The fourth is coverage of parameter-efficient fine-tuning heterogeneity. Every prior survey leaves at least one column unchecked, and the two camps fail on complementary ones. The from-scratch tradition misses the attenuation effect and PEFT heterogeneity. The foundation-model tradition misses the attenuation framing and the reconciliation. The present review sets out to be the first to address all four.

IX. Limitations and Scope

This work is a critical synthesis rather than a new empirical study. Two things bound the strength of our conclusions. The evidence we aggregate bounds it. The fact that we synthesize that evidence rather than generate it bounds it too. We make those bounds explicit here. For a venue such as TMLR, the scope of a claim is as load-bearing as the claim itself.

a) A qualitative alignment, not a pooled meta-analysis.: The confound audit (C1) asks whether reported heterogeneity-method gains survive a pretrained baseline. It does so by aligning findings *across* studies that were never designed to be compared head to head. The primary sources differ in modality, architecture, partitioning scheme, and even in how they define the heterogeneity "gap". Chen et al. [22] measure a FedAvg-minus-centralized gap over a CIFAR-10 Dirichlet (α=0.3) split, a by-city Cityscapes split, and a geographic iNaturalist split, and they mix top-line accuracy with mIoU. Nguyen et al. [21] measure an IID-versus-non-IID gap over CIFAR-10, FEMNIST, Stack Overflow, and Reddit, and they report it through figures rather than per-optimizer tables. Legate et al. [31] and Fanì et al. [32] evaluate frozen-feature recipes on yet other datasets and splits. The reference point, the skew mechanism [17, 18], and the metric all vary. So we deliberately do *not* pool effect sizes into a single number. The audit is a qualitative judgment about the *direction* and *survival* of effects. It is not a like-for-like re-benchmark. It also inherits whatever selection and reporting biases the underlying papers carry. No algorithm dominates across controlled partitions in this literature to begin with [19].

b) Reliance on preprint and single-case counter-evidence.: The boundary map (C2) stands on firmer ground where it draws on peer-reviewed results. Those results include LoRA aggregation instability [36], persistent classifier-head bias under label skew [44], normalization mismatch under covariate shift [42], and unseen-client shift [43]. It stands on softer ground where its sharpest claims come from preprints. Consider the most striking "amplification" observation. A pretrained DistilBERT+LoRA widens the worst-client accuracy *gap* relative to a from-scratch model at extreme skew. That observation comes from a non-peer-reviewed preprint [35]. So does the broader report that federated LoRA continues to struggle under the FedAvg paradigm [41]. We treat these as *suggestive* indicators of where heterogeneity may be relocated, not as settled findings. Our central conclusions do not rest on any single preprint. The LoRA-aggregation pathology we invoke is, in its clean form, an algebraic observation about a two-client equal-weight average [36]. Its extrapolation to many clients under weighted aggregation is plausible. The sources we cite do not establish it in full generality.

c) Uneven evidence across heterogeneity types.: Our strongest evidence is concentrated in *vision under label-distribution skew*. There, independent studies agree on three points. A pretrained start shrinks the heterogeneity gap. Studies measure that gap either against centralized training [22] or between IID and non-IID partitions [21]. A pretrained start also compresses the differences between federated optimizers [21]. And it leaves frozen-feature classifiers strongly competitive [31, 32]. Claims outside that regime are correspondingly more tentative. The language evidence is supportive but thinner. Even there a caveat applies. The source reports machine translation with pretrained models only as matching or beating centralized training. It does so because it provides no from-scratch MT baseline [33]. Evidence on *feature/covariate shift* [42, 43] and on the *federated-LLM and parameter-efficient* regime [36] is sparser still and spans fewer datasets. So our assertion that heterogeneity persists or relocates there is a hypothesis backed by suggestive evidence. It is not an established boundary. Two scope caveats sharpen this. First, the supporting theory covers a restricted model class. That class is two-layer CNNs. In those CNNs, heterogeneity acts through "misaligned filters" that a pretrained start partially avoids [23]. The theory does not speak directly to the deep networks and transformers used in practice. Second, most of the favorable vision evidence uses generic (ImageNet-style) pretraining. Yet in-domain self-supervised pretraining can outperform it under severe heterogeneity [45]. This suggests the benefit tracks domain alignment. The benefit need not transfer to settings where no well-aligned pretrained backbone exists.

d) No new experiments, and a falsification protocol we propose but do not execute.: Finally, this is a zero-experiment review. We run no controlled single-testbed audit of our own. So the unifying claim remains a synthesis-level hypothesis rather than a directly verified result. That claim holds that the *locus* of non-IID depends on the starting point and migrates into the aggregation operator and the classifier head. We offer the falsification protocol and research agenda (C3) precisely because existing benchmarks do not probe this axis

TABLE VI
COVERAGE OF THE TWO LITERATURES BY REPRESENTATIVE SURVEYS, AGAINST THE FOUR CAPABILITIES A RECONCILING SURVEY WOULD NEED. *From-scratch* MARKS A SURVEY THAT COVERS THE CLASSICAL FROM-SCRATCH NON-IID LITERATURE. *Attenuation* MARKS A SURVEY THAT COVERS THE PRETRAINED-INITIALIZATION ATTENUATION EFFECT (THE IID-VS-NON-IID GAP SHRINKING FOR FULL-MODEL FINE-TUNING UNDER LABEL SKEW ON IN-DISTRIBUTION METRICS). *Reconciles* MARKS A SURVEY THAT PLACES THE FROM-SCRATCH AND ATTENUATION EVIDENCE IN A SINGLE FRAME. *PEFT het.* MARKS A SURVEY THAT COVERS PARAMETER-EFFICIENT (E.G., LORA) FINE-TUNING HETEROGENEITY. "PARTIAL" = ACKNOWLEDGED WITHOUT DEDICATED TREATMENT. EVERY PRIOR SURVEY LEAVES AT LEAST ONE COLUMN UNCHECKED, AND THE TWO CAMPS FAIL ON COMPLEMENTARY ONES. CELLS REFLECT EACH SURVEY'S STATED SCOPE AS READ FROM ITS BODY TEXT. THE *Attenuation* AND *Reconciles* COLUMNS FOLLOW FROM THOSE WORKS TREATING HETEROGENEITY AS PERSISTING OR WORSENING WITHOUT ENGAGING THE PRETRAINED-INITIALIZATION EVIDENCE.

Survey	From-scratch	Attenuation	Reconciles	PEFT het.
Zhu et al. [24]	Yes	No	No	No
Kairouz et al. [3]	Yes	No	No	No
Ye et al. [25]	Yes	No	No	No
Jimenez Gutierrez et al. [26]	Yes	No	No	No
Ren et al. [27]	Partial	No	No	Partial
Li et al. [28]	Partial	No	No	Partial
Fan et al. [29]	Partial	No	No	Partial
Woisetschläger et al. [46]	Yes	Partial	No	No
This review	**Yes**	**Yes**	**Yes**	**Yes**

and prior heterogeneity taxonomies include no initialization category [25]. The protocol and agenda specify the controlled experiments that *would* confirm or refute the boundary map. We do not carry them out here. Our contribution is accordingly cartographic and agenda-setting. We reorganize existing evidence and specify the tests it calls for. We do not deliver a definitive measurement of where heterogeneity ultimately lives.

X. CONCLUSION

The non-IID problem organized a decade of federated learning. The field defined, measured, and "solved" it under one unexamined assumption. That assumption was training from scratch. Lifting the assumption does not make heterogeneity disappear. Instead, it relocates and re-ranks that heterogeneity. Full-model fine-tuning dissolves the label-skew accuracy gap. Heterogeneity remains, or worsens, in parameter-efficient fine-tuning, feature shift, and the worst-client tail. Reading the literature through initialization retires a class of questions. It also sharpens the ones that remain.

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